Provisional: regression outputs — reform-window DDD (DA15 + IDA15)

MTU15 thesis companion (design NOT pinned down)

Pablo Paramio

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Status. *This file collects the current reform-window DDD outputs. The identification design is being iterated — the descriptive foundations live in a sibling document and the framing of "heterogeneity vs treatment/control", covariate choice, and session-choice for the IDA outcomes is still in flux. Read the numbers here as the current state of play, not as final results.*

Contents

1 Reform-window identification (DA15 + IDA15)

The headline thesis question is the effect of **MTU15 granularity** on strategic bidding. Reforzada (the post-blackout RT2 pay-as-bid + zonal-monopoly windfall channel documented in the descriptive companion, Part B) is a parallel *confounding* mechanism — analogous in spirit to the operational/solar midday channel in the main paper (main paper §2.4): a real economic phenomenon that we want to characterise but *exclude from the identifying contrast* so as not to mix channels. This section's design picks reform windows in which reforzada is held constant, then differences out residual seasonality via a 2024 placebo.

Design. Two reform-windows, each chosen so that reforzada is constant within both the pre- and post-treatment sub-windows:

Table 1: Reform-window design. Reforzada is the post-blackout RT2 pay-as-bid regime (active 2025-04-28 onward). The 2024 placebo uses calendar-matched dates. PT assumption: YoY seasonal change in (crit − flat) was the same in 2024 as in 2025.

Reform / sample	Pre-window	Post-window	Reforzada	Length pre/post
IDA15 (2025)	2024-12-09 – 2025-03-18	2025-03-19 – 2025-04-27	off	100 / 40 d
IDA15 placebo (2024)	2023-12-09 – 2024-03-18	2024-03-19 – 2024-04-27	off	matching
DA15 (2025)	2025-04-28 – 2025-09-30	2025-10-01 – 2026-02-13	on	155 / 135 d
DA15 placebo (2024)	2024-04-28 – 2024-09-30	2024-10-01 – 2025-02-13	off	matching

The temporal design holds reforzada constant within each window (on for DA15, off for IDA15), and the 2024 placebo absorbs YoY seasonality that the within-day DiD alone could not. Each pre-window is ≥ 3 months; DA15 has 5 months post-reform.

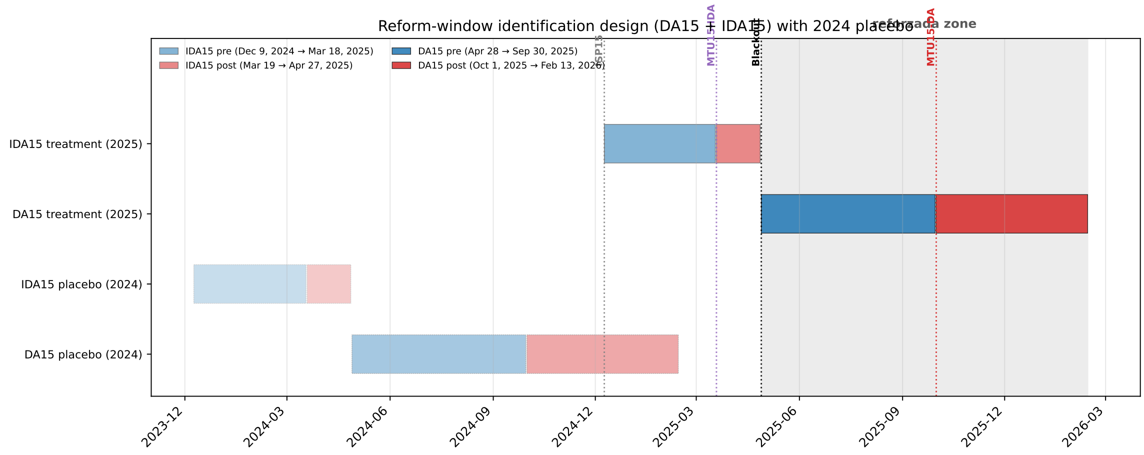

Figure 1: Timeline of reforms and DDD windows. Vertical guides: ISP15 (2024-12-09), MTU15-IDA (2025-03-19), Iberian blackout (2025-04-28), MTU15-DA (2025-10-01). Shaded grey: reforzada zone. Blue bars: pre-reform sub-windows; red bars: post-reform. 2024 placebo rows shown with lighter shading.

Triple-difference specification. Let $Y_{f,u,d,h}$ be the outcome at firm f, unit u, date d, clock-hour h. Define $\text{crit}_h = 1$ if $h \in \{5–9, 16–22\}$ (treated hour-class), restrict sample to $\{\text{crit}, \text{flat}\}$. Let $\text{post}_d = 1$ if d is after the reform date, $\text{y25}_d = 1$ if d falls in the 2025 (treatment) window. Then:

$$\begin{aligned} Y_{f,u,d,h} = \alpha &+ \beta_1 \text{crit}_h + \beta_2 \text{post}_d + \beta_3 \text{y25}_d + \beta_4(\text{crit} \times \text{post}) + \beta_5(\text{crit} \times \text{y25}) \\ &+ \beta_6(\text{post} \times \text{y25}) + \boxed{\beta_7(\text{crit} \times \text{post} \times \text{y25})} + \gamma_f + \mu_u + \delta_{\text{DOW}(d)} + \varepsilon_{f,u,d,h}. \end{aligned} \tag{1}$$

β_7 is the headline coefficient: the change in the (critical − flat) differential at the reform date, *net of* the corresponding 2024 seasonal change. FE: firm γ_f + unit μ_u + day-of-week δ_{DOW}. SE clustered at the date level. Equivalently $\beta_7 = \Delta\Delta Y^{2025} - \Delta\Delta Y^{2024}$ where $\Delta\Delta Y = (\bar{Y}_{\text{post,crit}} - \bar{Y}_{\text{post,flat}}) - (\bar{Y}_{\text{pre,crit}} - \bar{Y}_{\text{pre,flat}})$.

Event-study spec (visual pre-trends). For each outcome, plot the monthly differential separately for the 2025 treatment year and the 2024 placebo year, anchored to τ = months from the reform date:

$$\Delta Y_{\tau,y} \;=\; \bar{Y}_{\text{crit}}(\tau, y) - \bar{Y}_{\text{flat}}(\tau, y), \qquad y \in \{2024, 2025\}.$$

Identifying assumption test: for $\tau < 0$ the two series should overlap (parallel pre-trends); at $\tau \geq 0$ the 2025 series diverges from the 2024 series by exactly the treatment effect.

Identifying assumptions. *Formal PT (triple-difference).*

$$\mathbb{E}\big[\Delta\Delta Y(0) \mid \text{y25} = 1\big] \;=\; \mathbb{E}\big[\Delta\Delta Y(0) \mid \text{y25} = 0\big],$$

where $\Delta\Delta Y(0)$ is the untreated within-day pre/post change in the critical-flat differential.

Threats killed by the design.

- *Seasonality of the critical-flat spread* (summer-to-winter heating/AC patterns): differenced out by the 2024 placebo on the same calendar months.
- *Day-level cost shocks* (TTF gas, CO_2, demand spikes): within-day differencing on each date absorbs anything that hits both hour-classes proportionally.

- *Asymmetric gas-cost confound*: ruled out empirically in main paper §2.4 — CCGT price-setting share 9.1% crit vs 9.0% flat, so TTF shocks cannot propagate asymmetrically.
- *Operational midday channel* (solar drives prices down at midday): midday hours $\in \{11\text{–}14\}$ excluded from the control set; only flat hours $\in \{1, 2, 3\}$ enter as control.
- *Reforzada / RT2 pay-as-bid windfall*: held constant within each window by design (the central insight). Characterised as a confounder in the descriptive companion, Part B (REE post-clearing CCGT inclusion, geographic CCGT concentration, Cournot reading + CNMC sanctions).

Threats remaining.

- *Within-firm portfolio composition shifts*: addressed by unit FE; sensitivity check restricts to units active in both years.
- *Reform-anticipation effects*: tested by the event-study leads β_τ for $\tau \in \{-1, -2\}$.

What this DDD identifies (and what it does not). The within-day differencing on (crit − flat) absorbs day-level common shocks; the 2024 placebo absorbs YoY seasonal variation in the spread itself. Whether the identified parameter is the *differential* reform effect or the *absolute* effect on critical hours depends on whether flat hours are exposed to MTU15. This is a market-specific empirical question; the existing project evidence (descriptive companion, D_w decomposition section, DA Oct–Dec 2025) shows:

- **DA market**: flat hours show essentially zero within-hour granularity activity (GN, GE, HC: 0%–0.05% on both Δp and Δq channels; IB: 0.89%/1.03%, an order of magnitude below its critical-hour shaping). Critical hours, by contrast, show GN at 18.4% Δq-active. Flat hours are approximately MTU15-untreated for CCGT bidding. *The DA-cell DDDs on prices and on q_1 should therefore be read as close to the absolute critical-hour effect, not a within-day differential between two treated arms.*
- **IDA market** (pre-blackout, see descriptive companion): flat-hour cell activity is 7–21% across pivotal firms, against critical-hour 9–30%. Both hour-classes show within-hour activity in the IDA. *The IDA-cell DDD identifies a true differential, not an absolute effect on critical hours.*

Same logic for pivotal-vs-non-pivotal in the D_w decomposition: the table only breaks out the four pivotal firms because non-pivotal CCGT participation in the relevant zones is negligible. For granularity-driven outcomes the non-pivotal-as-control reading is more defensible than for aggregate outcomes like q_2, where the older 2-way DiD in the main paper's appendix shows non-pivotal firms do respond (with sign reversal). The framing is outcome- and market-specific: closer to treatment/control for DA-granularity-shaping outcomes, closer to heterogeneity for IDA and for aggregate q_2.

Heterogeneity decomposition by pivotal firm status. For firm-level outcomes (q_1, q_2) we report the DDD twice: pooled over the CCGT sample, and decomposed by pivotal status (Big-4: IB, GE, GN, HC, EDP-PT; non-pivotal: Repsol, Engie, TotalEnergies, Moeve). The decomposition is a *heterogeneity analysis*, not a fourth difference: non-pivotal CCGT firms are exposed to the same MTU15 reform as pivotal firms and the older 2-way DiD in the main paper's appendix in fact shows them responding (the q_2 sign on (crit × post) reverses between pivotal and non-pivotal subsamples). We do not assume they are an "untreated control"; we read the contrast as how the reform's effect varies with market power. Operationally this is implemented as a saturated OLS-FE regression with all crit/post/y25/pivotal interactions (the pivotal main effect is firm-invariant and absorbed by γ_f), yielding a coefficient on (crit $\times$ post $\times$ y25 $\times$ piv) that is mathematically the difference of the two sub-sample DDDs. We label it δ_{piv} rather than β_{1234} to make the heterogeneity-not-fourth-difference reading explicit:

$$\delta_{\text{piv}} \equiv \widehat{\text{DDD}}_{\text{pivotal}} - \widehat{\text{DDD}}_{\text{non-pivotal}}.$$

SE clustered at the firm level (pivotal does not vary within firm; $G \approx 9$ — wild-cluster bootstrap as a robustness step is flagged below). The DDDD-style name and the "quadruple difference" framing previously used in this section have been retired: in light of the credibility-quadratic-in-differences argument (Roth, paraphrased in Borusyak's lecture notes), and given the empirical evidence that non-pivotal firms are not a clean control, the heterogeneity-decomposition interpretation is the only one we sustain.

Outcomes and joint q_1/q_2 sign-quadrants. Two firm-level quantity outcomes and two market-level price outcomes. The firm-level pair is informative *only when read jointly*: q_1 (DA-cleared MWh) and q_2 (signed intraday MWh) jointly characterise the firm's strategic posture, with each $(\text{sign}(\beta_{q_1}), \text{sign}(\beta_{q_2}))$ quadrant mapping to a distinct economic narrative. Table 2 lays this out. D_w, Δp, Δq are *not* DDD outcomes because they require within-hour quarters that do not exist pre-MTU15; they appear as post-only cross-section in the descriptive companion (D_w decomposition and D_w distribution sections).

Table 2: Joint (q_1, q_2) sign quadrants and the firm-behaviour narrative each implies. The DDD coefficient on critical-flat-spread for each quantity outcome gives the row/column sign. Each cell is a coherent story about how dispatchable firms responded to the reform; only one is realised in the data.

	$\beta(q_1) > 0$ (more DA-cleared)	$\beta(q_1) < 0$ (less DA-cleared)
$\beta(q_2) > 0$ (more IDA selling)	*Production expansion*: firms scale up across both stages (unlikely under unchanged demand).	*Ito–Reguant withholding*: classic strategic relabelling — withhold in DA to drive up clearing, sell volume back in IDA.
$\beta(q_2) < 0$ (less IDA selling)	*Operational refinement*: granular DA absorbs within-hour variation that previously needed IDA repositioning; auctions clear better, less intraday correction needed.	*Broader disengagement*: firms reduce participation in *both* auction-cleared stages, with the outside option being REE post-clearing redispatch (RT2 pay-as-bid). Not Ito–Reguant.

For prices, the within-day spread response separately tests the operational-refinement vs strategic-exploitation directions: negative ($\beta_7 < 0$, spread compresses) if granularity lets non-pivotal competitors fill critical-hour 15-min slots; positive ($\beta_7 > 0$, spread widens) if pivotal firms exploit the new pricing dimension.

How to read Figure 2. Each panel is a single (outcome $\times$ reform) cell. The horizontal axis is τ = months relative to the reform date (e.g. $\tau = 0$ is March 2025 for IDA15 and October 2025 for DA15). The vertical axis is the monthly mean of the within-day differential $\bar{Y}_{\text{crit}}(\tau) - \bar{Y}_{\text{flat}}(\tau)$ in that month — a scalar that tells us how much higher the critical-hour outcome is relative to the same day's flat-hour outcome.

Each panel has two coloured lines:

- Red (2025): the actual treatment year. At $\tau = 0$, MTU15 is implemented.
- Blue (2024): the calendar-matched placebo. At $\tau = 0$, the analogous calendar month one year earlier — no reform, just seasonality.

The DDD coefficient β_7 is a difference of *changes* (not levels). In raw form the two lines can sit at different levels (e.g., 2025 prices systematically higher than 2024 because of inflation) and the design still works — the level shift is absorbed by the y25 main effect β_3 and what matters is that the (red $-$ blue) gap stays constant pre-reform and shifts post-reform.

The figure normalises each line by subtracting its value at $\tau = -1$, so both start at zero at the omitted reference month. After normalisation the visual test is the standard event-study form:

- For $\tau < 0$ (pre-reform): red and blue should **overlap** (parallel-trends test).
- At $\tau \geq 0$ (post-reform): a persistent vertical separation between red and blue is the treatment effect.
- β_7 is the average post-period vertical separation between the two normalised lines.

Figure 2: Event-study (reduced-form, normalised). Rows = outcomes (DA price, IDA price, q_1, q_2); columns = reforms (IDA15, DA15). Each panel plots the monthly (critical $-$ flat) within-day differential, anchored to τ = months from the reform date and *normalised by subtracting the* $\tau = -1$ *value*, so both lines start at 0 at $\tau = -1$. Red = 2025 (treatment); blue = 2024 (placebo). Pre/post regions lightly shaded. Reduced-form monthly means; full regression-based event-study with CI follows the headline DDD.

Pre-trend reading. DA15: pre-period overlap is reasonable for prices; q_1 shows the 2025 series sitting below the 2024 series *already before* $\tau = 0$, signalling that the reforzada channel pre-bends the differential before MTU15-DA fires — a partial PT violation for q_1 on this window. IDA15: pre-period overlap looks reasonable for prices; post-period is too short to read visually. The headline DDD with date-clustered SE follows.

Estimators in practice: OLS-FE and doubly-robust DiD on prices in levels. We report two estimators for the same DDD parameter, each with three to four specifications. Both use clearing prices in EUR/MWh as the outcome (in levels), and both target the same coefficient: the (critical $-$ flat) differential response to the reform, net of the 2024 calendar placebo.

Panel construction. For each (reform window, market) combination $w \in \{\text{DA15}, \text{IDA15}\} \times \{\text{DA}, \text{IDA-S1}\}$ we build two analytic panels:

- *Period panel*: one row per (date, ISP-period) tuple. Pre-MTU15 days contribute 1 observation per clock-hour (MTU60); post-MTU15 days contribute 4 (MTU15). Sample restricted to clock-hours in {critical ∪ flat}. Used for the OLS-FE estimator. Sizes: 13 888 obs (DA15-DA), 19 138 obs (DA15-IDA), 3 934 obs (IDA15-DA), 5 530 obs (IDA15-IDA).

- *Spread panel*: one row per date with the within-day means $\bar{Y}_d^{\text{crit}} = \frac{1}{|H_{\text{crit}}|} \sum_{p \in H_{\text{crit}}} Y_{d,p}$ and $\bar{Y}_d^{\text{flat}}$ as two columns. Used for the DR-DiD estimator (one drdid call per hour-class). Sizes: 584 dates (DA15-DA), 554 (DA15-IDA), 281 (IDA15-DA), 278 (IDA15-IDA).

OLS-FE DDD. On the period panel, three specifications, each with the same DDD coefficient β_7 but different fixed-effect structures:

$$
\begin{aligned}
\text{(M1) sparse:}\quad & Y_{d,p} = \alpha + \sum_{k=1}^{7} \beta_k Z_k + \gamma_h + \delta_{\text{DOW}(d)} + \varepsilon_{d,p}, \\
\text{(M2) month FE:}\quad & Y_{d,p} = \alpha + \sum_{k=1}^{7} \beta_k Z_k + \gamma_h + \delta_{\text{DOW}(d)} + \eta_{\text{month}(d)} + \varepsilon_{d,p}, \\
\text{(M3) date FE:}\quad & Y_{d,p} = \alpha + \beta_1 \text{crit} + \beta_4(\text{crit}\times\text{post}) + \beta_5(\text{crit}\times\text{y25}) \\
& \qquad + \beta_7(\text{crit}\times\text{post}\times\text{y25}) + \gamma_h + \delta_{\text{DOW}(d)} + \nu_d + \varepsilon_{d,p}.
\end{aligned}
$$

Here $(Z_1, \ldots, Z_7) = (\text{crit}, \text{post}, \text{y25}, \text{crit}\times\text{post}, \text{crit}\times\text{y25}, \text{post}\times\text{y25}, \text{crit}\times\text{post}\times\text{y25})$, γ_h is a clock-hour FE, $\delta_{\text{DOW}(d)}$ is a day-of-week FE, $\eta_{\text{month}(d)}$ is a calendar-month FE, and ν_d is a date FE that absorbs all date-level terms in M3 (so the date-level $\beta_2, \beta_3, \beta_6$ drop out and β_7 is identified purely from within-date variation). SE clustered at the date level via `reghdfe ..., vce(cluster date)`.

Doubly-robust DiD (Sant'Anna–Zhao 2020). On the spread panel, two 2×2 doubly-robust DiDs per (window, market), one on each hour-class outcome:

$$
\widehat{\text{ATT}}_c^{\text{DR}} = \mathbb{E}_n\Big[\Big(w_1(D) - w_0(D, X; \hat{p})\Big)\Big((Y_{\text{post}}^c - Y_{\text{pre}}^c) - \big(\hat{m}_{\text{post}}^0(X) - \hat{m}_{\text{pre}}^0(X)\big)\Big)\Big],
$$

with $c \in \{\text{crit}, \text{flat}\}$, treatment indicator $D = \text{y25}_d$, time indicator $t = \text{post}_d$, covariates $X = (\text{i.dow}, \text{i.month})$, and

$$
w_1(D) = \frac{D}{\mathbb{E}_n[D]}, \qquad w_0(D, X; \hat{p}) = \frac{\hat{p}(X)(1-D)/(1-\hat{p}(X))}{\mathbb{E}_n[\hat{p}(X)(1-D)/(1-\hat{p}(X))]}.
$$

The default `drdid` estimator (`drimp`) uses inverse-probability-tilting for $\hat{p}(X)$ and weighted least squares for the outcome model $\hat{m}^0(X)$. Three nuisance-model specifications are reported:

- *DR1*: no covariates ($X = \emptyset$). Reduces to the unconditional 2×2 DiD on the spread Y^c.

- *DR2*: $X = (\text{i.dow}, \text{i.month})$, asymptotic SE via the influence function.

- *DR3*: $X = (\text{i.dow}, \text{i.month})$, wild-bootstrap inference (999 Mammen draws, `drdid ..., wboot rseed(4242)`).

DDD via the difference of two doubly-robust DiDs (the headline). The triple difference is the difference of the two hour-class ATTs:

$$
\widehat{\text{DDD}} = \widehat{\text{ATT}}_{\text{crit}}^{\text{DR}} - \widehat{\text{ATT}}_{\text{flat}}^{\text{DR}}.
$$

Joint inference uses the recentered influence functions $\hat{\psi}_i^c$ saved by drdid's `stub(`c`)` option, which satisfy $\sqrt{N}(\widehat{\text{ATT}}^c - \text{ATT}^c) = N^{-1/2} \sum_i \hat{\psi}_i^c + o_p(1)$ (Sant'Anna–Zhao 2020, Theorem 1). Because the crit and flat means are computed on the same dates (within-date dependence), the variance of the DDD is taken from the variance of the IF *difference*:

$$\widehat{\mathrm{Var}}(\widehat{\mathrm{DDD}}) \;=\; \frac{1}{N}\,\widehat{\mathrm{Var}}_n\Big[\hat{\psi}_i^{\mathrm{crit}} - \hat{\psi}_i^{\mathrm{flat}}\Big], \qquad N = \text{number of dates},$$

i.e. the cross-class covariance is captured automatically by the joint distribution of IFs. The two-sided p-value is reported under asymptotic normality.

Why two estimators. The OLS-FE DDD targets the standard triple-interaction coefficient β_7 under the unconditional PT assumption $\mathbb{E}[\Delta\Delta Y(0) \mid \mathrm{y25}=1] = \mathbb{E}[\Delta\Delta Y(0) \mid \mathrm{y25}=0]$. The DR-DiD targets the conditional ATT under the conditional PT $\mathbb{E}[Y_{\mathrm{post}}(0) - Y_{\mathrm{pre}}(0) \mid \mathrm{y25}, X] = \mathbb{E}[Y_{\mathrm{post}}(0) - Y_{\mathrm{pre}}(0) \mid X]$, and is semiparametrically efficient under correct specification of the nuisance models (Sant'Anna–Zhao 2020). The two estimands coincide when X is balanced across treatment cells; with calendar-matched windows, balance on $X = (\mathrm{dow}, \mathrm{month})$ is approximate but not exact (a year-on-year drift of ≈ 1 day-of-week, plus daylight-saving boundaries). The discrepancy between the OLS-FE and DR-DiD point estimates is therefore the IPW contribution.

Naive before-after vs DDD: how much of the apparent reform effect is seasonal bias? The standard alternative to a DDD design is a within-year before-after (BA) regression with calendar (and sometimes weather) controls. Table 3 compares the BA-with-calendar-controls estimate for each of our four (window, market) cells against the DDD with the same controls. The seasonal bias — what the BA misattributes to the reform but the DDD nets out via the 2024 placebo — is striking on the DA15 window: BA gives +26 EUR/MWh on the critical-flat spread, DDD gives essentially zero. The 2024 calendar-matched placebo year shows the same +27 swing in its own pre-vs-post comparison; almost the entire apparent BA effect is seasonal swing of the critical-flat spread from summer to winter, not the MTU15 reform. On the IDA15 window the bias is smaller (3–5 EUR/MWh) but goes the *opposite direction* (the 2024 placebo shows a ≈ -8 EUR/MWh swing, vs +2/+6 in the 2025 treatment year). In both windows the BA is uninformative about the reform; the DDD is the design that isolates the reform from the calendar.

Table 3: Naive before-after (BA) vs DDD on the within-day (crit−flat) spread, EUR/MWh, all four (window, market) cells. BA: within-year (post − pre) on 2025 spread, OLS with `i.dow+i.month` calendar controls and date-clustered SE. Placebo BA: same spec on 2024 spread. DDD: 2×2 DiD with both years and same calendar controls (the headline OLS-FE coefficient on $\beta_{\mathrm{post}\times\mathrm{y25}}$). *Seasonal bias* = BA − DDD = what an advisor's naive within-year BA-with-controls attributes to the reform but is really seasonality (identified by the 2024 placebo). Standard errors in parentheses. Stars: * 0.10, ** 0.05, *** 0.01.

Window	Outcome	BA 2025 (a)	BA 2024 (placebo) (b)	DDD (c)	Seasonal bias (a) − (c)
DA15	DA price	25.83*** (3.72)	26.99*** (4.34)	-0.17 (2.41)	26.00
DA15	IDA-S1 price	25.90*** (3.79)	26.42*** (3.99)	-2.24 (2.33)	28.15
IDA15	DA price	2.24 (5.30)	-8.49*** (3.21)	-1.74 (3.35)	3.98
IDA15	IDA-S1 price	6.07 (4.91)	-7.05** (3.25)	0.74 (3.31)	5.33

Headline DDD estimates: clearing prices, all (reform window, outcome) cells. Every price outcome is estimated in *both* reform windows. The natural mapping (DA price tested in DA15 window, IDA price tested in IDA15 window) is the direct-effect cell; the off-diagonal cells (DA price in IDA15 window, IDA price in DA15 window) test *spillover*. We report the doubly-robust DiD (Sant'Anna–Zhao 2020, `drdid`) on prices in levels, decomposed into $\mathrm{ATT}_{\mathrm{crit}}$ and $\mathrm{ATT}_{\mathrm{flat}}$, with the DDD given by the difference. Joint SE for the DDD comes from the difference of the recentered influence functions (drdid `stub()` option), which correctly handles the within-date covariance between the two hour-class means. Covariates: `i.dow+i.month`. Full per-cell OLS-FE companion tables: Tables 5–8.

Table 4: Headline DDD on *prices in levels*, EUR/MWh, all four (reform window, market) cells. DR-DiD columns (Sant'Anna–Zhao 2020, drimp): ATT_{crit} and ATT_{flat} are 2×2 DiDs per hour-class with covariates `i.dow+i.month`; DDD is their difference with joint SE from the influence-function difference. OLS-FE β_7 column: triple-interaction coefficient from `reghdfe` on the period-level panel with clock-hour + DOW + month FE, SE clustered at date. Standard errors in parentheses. Stars: * 0.10, ** 0.05, *** 0.01. The *Channel* column distinguishes direct (the market hit by the reform) from spillover (the cross-market test): DA15-DA and IDA15-IDA are direct; the other two cells are spillover.

		DR-DiD on levels			OLS-FE
Window	Outcome	ATT_{crit}	ATT_{flat}	DDD	β_7
DA15	DA price	-8.34^{*} (4.33)	-2.96 (4.85)	-5.38^{**} (2.36)	-0.29 (2.92)
DA15	IDA-S1 price	-10.38^{**} (4.46)	-5.02 (4.96)	-5.36^{**} (2.30)	-2.67 (2.85)
IDA15	DA price	-25.02^{***} (5.49)	-31.30^{***} (6.16)	$+6.28^{**}$ (3.01)	-2.24 (3.63)
IDA15	IDA-S1 price	-25.40^{***} (5.59)	-34.82^{***} (6.13)	$+9.43^{***}$ (2.95)	$+0.50$ (3.69)

OLS-FE vs DR-DiD. Both estimators target the DDD parameter; the gap between them is the IPW reweighting on calendar covariates. Without covariates, DR-DiD reproduces OLS-FE to 3 decimal places (sanity check). With $X = (\text{i.dow}, \text{i.month})$, DR-DiD reweights observations so the (dow, month) distribution is balanced across 2024-placebo and 2025-treatment cells before computing the ATT. This balance is approximate but not exact under calendar-matched windows (day-of-week drifts ≈ 1 position across years; DST boundaries differ by a day). The OLS-FE estimand is the unconditional uniform-weight DDD; the DR-DiD estimand is the conditional ATT under conditional PT. We report both. The DR-DiD is the preferred headline only insofar as conditional PT given $(\text{dow}, \text{month})$ is the assumption we are willing to maintain — a discussion of covariate adequacy is below.

Reading the four price cells. DA15: both DA and IDA Session 1 spreads compress by ≈ 5.4 EUR/MWh after MTU15-DA. Same direction, near-identical magnitude across markets (−5.38, −5.36). The IDA-cell is a clean spillover because MTU15-IDA is held constant within the DA15 window in both years. IDA15: both spreads widen (+6.3 DA, +9.4 IDA), again same direction across markets, with the directly-affected IDA larger. Hour-class decomposition: in DA15, critical and flat both fall (−3 to −10 EUR/MWh) with critical falling more; in IDA15, both fall sharply (−25 to −35, post-MTU15-IDA is structurally cheaper) with flat falling more. The DDD identifies the differential of these moves, not the absolute response of critical hours. Same-direction-within-window rules out a market-specific artefact; cross-window asymmetry (DA15 compresses, IDA15 widens) is a property of the reforms, not of the markets being measured.

Mechanism reading (pending joint q_1/q_2 check). The price DDDs are consistent with two compositional stories that we cannot separate from prices alone: (a) MTU15-DA opens 4× finer critical-hour quarter-slots that non-pivotal generators (renewables, hydro, fast-ramp gas) fill while pivotal firms reduce DA participation, compressing the spread; (b) MTU15-IDA reshapes IDA bidding so that the post-reform IDA market is structurally different in ways the DDD only captures via spread widening. The firm-level q_1/q_2 regressions below are the test that pins down which compositional story is consistent with the data.

Follow-ups flagged in priority order: (a) BA-with-controls comparison to quantify how much of the apparent reform effect the DDD absorbs as seasonal; (b) closest-to-delivery IDA price (in place of Session 1) as a robustness for the IDA-spillover cell, given the June 2024 IDA reform redefined Session 1; (c) augmented DR-DiD covariate set with ENTSO-E wind+solar generation and TTF gas as exogenous structural cost shifters; (d) propensity-score overlap diagnostic to confirm DR-DiD weights are not extreme; (e) event-study with multiplier-bootstrap simultaneous CI.

Table 5: OLS-FE DDD: DA clearing price in DA15 window. SE clustered at date.

	(1)		(2)		(3)	
	Sparse		Month FE		Date FE	
	b	se	b	se	b	se
crit*post*y25 (β_7, DDD)	-0.286	(2.915)	-0.286	(2.916)	-0.286	(2.915)
crit*post	28.057***	(2.073)	28.057***	(2.074)	28.057***	(2.073)
crit*y25	-3.529*	(1.951)	-3.529*	(1.952)	-3.529*	(1.951)
post*y25	-26.523***	(5.983)	-26.523***	(5.335)		
crit	0.000	(.)	0.000	(.)	0.000	(.)
post	11.846***	(4.112)	0.000	(.)		
y25	-1.662	(4.010)	-1.662	(2.947)		
Observations	13888		13888		13888	
R^2	0.282		0.416		0.800	
Clock-hour FE	Yes		Yes		Yes	
DOW FE	Yes		Yes		Yes	
Month FE	No		Yes		n/a	
Date FE	No		No		Yes	
Cluster	Date		Date		Date	

Table 6: OLS-FE DDD: IDA Session 1 clearing price in DA15 window. SE clustered at date.

	(1)		(2)		(3)	
	Sparse		Month FE		Date FE	
	b	se	b	se	b	se
crit*post*y25 (β_7, DDD)	-2.674	(2.852)	-2.674	(2.853)	-2.674	(2.852)
crit*post	28.703***	(1.987)	28.703***	(1.988)	28.703***	(1.987)
crit*y25	-2.421	(1.958)	-2.421	(1.958)	-2.421	(1.958)
post*y25	-27.432***	(6.030)	-26.886***	(5.313)		
crit	0.000	(.)	0.000	(.)	0.000	(.)
post	11.870***	(4.038)	0.000	(.)		
y25	-3.158	(4.075)	-3.210	(2.961)		
Observations	19138		19138		19138	
R^2	0.299		0.479		0.785	
Clock-hour FE	Yes		Yes		Yes	
DOW FE	Yes		Yes		Yes	
Month FE	No		Yes		n/a	
Date FE	No		No		Yes	
Cluster	Date		Date		Date	

Table 7: OLS-FE DDD: DA clearing price in IDA15 window. SE clustered at date.

	(1) Sparse		(2) Month FE		(3) Date FE	
	b	se	b	se	b	se
crit*post*y25 (β_7, DDD)	-2.243	(3.630)	-2.243	(3.631)	-2.243	(3.628)
crit*post	-10.440***	(1.738)	-10.440***	(1.739)	-10.440***	(1.738)
crit*y25	2.657	(2.282)	2.657	(2.284)	2.657	(2.282)
post*y25	-27.426***	(6.459)	-27.438***	(6.250)		
crit	0.000	(.)	0.000	(.)	0.000	(.)
post	-34.767***	(4.266)	-3.141	(6.004)		
y25	45.676***	(4.598)	45.670***	(4.229)		
Observations	3934		3934		3934	
R^2	0.573		0.627		0.874	
Clock-hour FE	Yes		Yes		Yes	
DOW FE	Yes		Yes		Yes	
Month FE	No		Yes		n/a	
Date FE	No		No		Yes	
Cluster	Date		Date		Date	

Table 8: OLS-FE DDD: IDA Session 1 clearing price in IDA15 window. SE clustered at date.

	(1) Sparse		(2) Month FE		(3) Date FE	
	b	se	b	se	b	se
crit*post*y25 (β_7, DDD)	0.504	(3.694)	0.504	(3.695)	0.504	(3.693)
crit*post	-9.933***	(1.693)	-9.933***	(1.694)	-9.933***	(1.693)
crit*y25	3.266	(2.155)	3.266	(2.155)	3.266	(2.154)
post*y25	-29.831***	(6.566)	-29.975***	(6.342)		
crit	0.000	(.)	0.000	(.)	0.000	(.)
post	-36.304***	(4.397)	-6.630	(5.983)		
y25	45.197***	(4.585)	45.316***	(4.202)		
Observations	5530		5530		5530	
R^2	0.588		0.627		0.837	
Clock-hour FE	Yes		Yes		Yes	
DOW FE	Yes		Yes		Yes	
Month FE	No		Yes		n/a	
Date FE	No		No		Yes	
Cluster	Date		Date		Date	

Table 9: DR-DiD on DA clearing price in DA15 window, prices in levels (Sant'Anna–Zhao 2020). 2×2 DiD per hour-class, covariates `i.dow + i.month`; DDD is the difference of ATTs with joint SE from the influence-function difference.

Coefficient	Estimate	SE
ATT_{crit}	-8.336	(4.326)
ATT_{flat}	-2.961	(4.846)
DDD = $ATT_{crit} - ATT_{flat}$	-5.375	(2.356)
p-value (DDD)	0.023	
95% CI (DDD)	[-9.992, -0.758]	
N (dates)	584	
Outcome	DA clearing price	
Window	DA15 (reform = 2025-10-01)	

Table 10: DR-DiD on IDA Session 1 clearing price in DA15 window, prices in levels (Sant'Anna–Zhao 2020). 2×2 DiD per hour-class, covariates `i.dow + i.month`; DDD is the difference of ATTs with joint SE from the influence-function difference.

Coefficient	Estimate	SE
ATT_{crit}	-10.379	(4.463)
ATT_{flat}	-5.016	(4.957)
DDD = $ATT_{crit} - ATT_{flat}$	-5.364	(2.296)
p-value (DDD)	0.019	
95% CI (DDD)	[-9.864, -0.863]	
N (dates)	554	
Outcome	IDA Session 1 clearing price	
Window	DA15 (reform = 2025-10-01)	

Table 11: DR-DiD on DA clearing price in IDA15 window, prices in levels (Sant'Anna–Zhao 2020). 2×2 DiD per hour-class, covariates `i.dow + i.month`; DDD is the difference of ATTs with joint SE from the influence-function difference.

Coefficient	Estimate	SE
ATT_{crit}	-25.019	(5.486)
ATT_{flat}	-31.302	(6.161)
$DDD = ATT_{crit} - ATT_{flat}$	6.283	(3.009)
p-value (DDD)	0.037	
95% CI (DDD)	[0.386, 12.180]	
N (dates)	281	
Outcome	DA clearing price	
Window	IDA15 (reform = 2025-03-19)	

Table 12: DR-DiD on IDA Session 1 clearing price in IDA15 window, prices in levels (Sant'Anna–Zhao 2020). 2×2 DiD per hour-class, covariates `i.dow + i.month`; DDD is the difference of ATTs with joint SE from the influence-function difference.

Coefficient	Estimate	SE
ATT_{crit}	-25.396	(5.587)
ATT_{flat}	-34.823	(6.129)
$DDD = ATT_{crit} - ATT_{flat}$	9.427	(2.949)
p-value (DDD)	0.001	
95% CI (DDD)	[3.647, 15.207]	
N (dates)	278	
Outcome	IDA Session 1 clearing price	
Window	IDA15 (reform = 2025-03-19)	

IDA-price robustness: closest-to-delivery session. The IDA-cell DDDs in Table 4 use Session 1 price throughout. Session 1's clearing horizon was redefined at the SIDC reform of June 2024 (MIBEL 6-session → SIDC 3-session, per the project's Spanish-market reference §IDA): under MIBEL Session 1 closed at 14:00 D−1 and targeted all 24 hours of D; under SIDC it covers all 96 ISP15 periods of D. Both target the same horizon nominally, but liquidity, participants, and the SIDC coupling regime differ in ways the within-window pre/post differencing cannot absorb when the placebo year is MIBEL and the treatment year is SIDC. As a robustness, we re-estimate the IDA-cell DR-DiD using the *closest-to-delivery* session price per (date, period) — the row with the maximum `session_number`, which under SIDC is IDA-3 for afternoons and IDA-2 for mornings, and under MIBEL is Session 4–6 for D-day afternoons and Session 3 for mornings. This is MTU- and session-count-invariant by construction. Results in Table 13:

Table 13: DR-DiD on IDA prices: Session 1 (baseline, in Table 4) vs closest-to-delivery session (robustness). For each (date, period) the closest-to-delivery price is the row with the maximum `session_number`: under SIDC, IDA-3 for afternoon and IDA-2 for morning; under MIBEL, Sessions 4–6 for D-day afternoons and Session 3 for morning. This robustness removes the MIBEL-to-SIDC redefinition of Session 1. Covariates: `i.dow+i.month`. Standard errors in parentheses. Stars: * 0.10, ** 0.05, *** 0.01.

Window	ATT_{crit}	ATT_{flat}	DDD
DA15	-2.54 (4.39)	-3.76 (4.75)	1.21 (2.33)
IDA15	-27.84 (5.16)	-34.10 (5.88)	6.26* (3.38)

The DA15-IDA "spillover" cell — baseline DDD of -5.36 EUR/MWh (significant) — effectively disappears under the robustness ($+1.21$, $p = 0.60$). The IDA15-IDA "direct" cell shrinks from $+9.43^{***}$ to $+6.26$ ($p = 0.064$), losing conventional significance. **The two IDA-side DDD numbers in the headline table are partly artefacts of the Session 1 choice**: the DA15 spillover is not robust at all, and the IDA15 direct effect is robust in direction and order of magnitude but loses statistical significance. The DA-side cells of the headline table (DA15-DA and IDA15-DA) are unaffected by this robustness (no session choice in the DA market). We will treat the IDA-side numbers in the headline as upper bounds on the true DDD until the wild-cluster-bootstrap and closest-to-delivery results converge.

Propensity-score overlap diagnostic. The DR-DiD propensity score $\hat{p}(X) = \Pr(\text{y25} = 1 \mid \text{i.dow}, \text{i.month})$ is estimated by logit on each (window, market) spread panel; weights blow up if $\hat{p}(X)$ clusters near 0 or 1. Across all four cells, $\hat{p}(X)$ ranges $[0.38, 0.54]$ (DA15-IDA) to $[0.49, 0.51]$ (DA15-DA), with median ≈ 0.50 in every cell and 0% of observations below 0.10 or above 0.90. Calendar-matched windows deliver essentially balanced treatment assignment in (dow, month) cells by design, so the IPW weights are not extreme and the DR-DiD inference is not driven by propensity-score tail behaviour. Per-cell overlap histograms are produced as working figures.

Headline DDD estimates: firm-level outcomes (q_1, q_2). Outcomes at the firm level: q_1 = DA-cleared MWh per (firm, unit, date, clock-hour) from OMIE `pdbc`, q_2 = signed intraday programmed MWh from `pibci`, both zero-filled for offline cells. CCGT-only sample: 42 pivotal Big-4 units (IB, GE, GN, HC, EDP-PT) and 8 non-pivotal units (Repsol, Engie, TotalEnergies, Moeve); untagged firms dropped. We report the pooled DDD (β_7) and the pivotal-status heterogeneity contrast $\delta_{\text{piv}} = \widehat{\text{DDD}}_{\text{pivotal}} - \widehat{\text{DDD}}_{\text{non-pivotal}}$ (per the framing above; this is what we previously over-labelled β_{1234}). All specs include firm + unit + clock-hour + DOW fixed effects. Full per-cell tables: Tables 15–18.

Table 14: Pooled DDD (β_7) and pivotal-status heterogeneity contrast (δ_{piv}) on firm-level q_1 and q_2, MWh per unit-hour. $\delta_{\text{piv}} > 0$ means pivotal firms' DDD effect is *larger* (in algebraic terms) than non-pivotal firms'. SE clustered at date (pooled) and at firm (decomposed; $G \approx 9$, asymptotic SE may under-cover, wild-cluster bootstrap recommended). Standard errors in parentheses. Stars: * 0.10, ** 0.05, *** 0.01.

		Pooled DDD (β_7)		δ_{piv} (heterogeneity)	
Outcome	Window	β_7	SE	δ_{piv}	SE
q_1	DA15	-16.27^{***}	(2.69)	-18.65^{**}	(7.35)
q_1	IDA15	-9.96^{***}	(3.10)	-4.01	(6.55)
q_2	DA15	-1.27^{**}	(0.64)	$+18.78$	(15.82)
q_2	IDA15	-1.70^{**}	(0.85)	$+11.10$	(10.81)

Joint (q_1, q_2) *reading.* Locating each window's pooled β_7 on the Table 2 sign-matrix: DA15 lands in ($\beta(q_1) < 0$, $\beta(q_2) < 0$) and IDA15 lands in the same quadrant. Both reforms produce **broader disengagement**, not Ito–Reguant withholding (which would require $\beta(q_2) > 0$): dispatchable CCGT firms reduce both DA-cleared volume and intraday programmed volume. The natural reading is that the relevant outside option for these units is not the IDA leg but REE's post-clearing redispatch (RT2 pay-as-bid; characterised in the descriptive companion, Part B). This is a substantively different story from the classical Ito–Reguant signature.

Pivotal-status heterogeneity reading. (i) For q_1 in the DA15 window the pooled DDD effect is sharply concentrated in pivotal firms: $\delta_{\text{piv}} = -18.65$ MWh/unit-hour, with the pooled β_7 (-16.27) entirely absorbed by the heterogeneity contrast. Non-pivotal firms in this window show no measurable change. (ii) For q_1 in the IDA15 window the pooled effect is significant (-9.96) but δ_{piv} is small and non-significant: the IDA reform pulled all CCGT firms in the same direction, no market-power moderation. (iii) For q_2, δ_{piv} has wide SEs and is not identifiable in either window. (iv) These contrasts identify heterogeneity *in the DDD effect*, not the absolute effect on pivotal firms — they tell us where the reform's response is concentrated, not whether the response is "strategic" in a structural sense.

Tension with the older 2-way DiD in `paper.tex`*.* The appendix in `paper.tex` (commented-out main table) reports a 2-way DiD (no 2024 placebo) on q_2 in pivotal vs non-pivotal subsamples with *opposite signs* (+3.87 pivotal vs −4.74 non-pivotal, both highly significant). Our DDD on the same outcome shows both subsamples moving in the *same* direction (both negative). The sign reversal in the older 2-way DiD was driven by the seasonality that the 2024 placebo here absorbs. Reconciling: once we net out YoY seasonality, the pivotal-vs-non-pivotal contrast in q_2 is wide-SE and uninformative — the headline pattern in the appendix was partly seasonal.

Caveats. (a) Firm-cluster $G \approx 9$ is small; asymptotic SE may under-cover. Wild-cluster bootstrap (`boottest`) is a planned robustness. (b) Pivotal/non-pivotal partition treats all 5 Big-4 firms as identical and all 4 placebo firms as identical; firm-by-firm decomposition is a natural next step (Moeve has a documented pre-trend break in the main paper that may pull the non-pivotal subsample). (c) The 2024 placebo for the DA15 q_1 panel includes 6 weeks (Apr 28 – Jun 13, 2024) before the June 14, 2024 IDA reform changed sessions 6 → 3. Robustness fix: use closest-to-delivery IDA price (MTU- and session-count-invariant) instead of Session 1 for the IDA-spillover regressions.

Granularity strategy. The cross-reform DDD runs on disaggregated period-level data, not on hourly aggregates. Pre-MTU15 the period is a 60-minute slot (1 obs / hour); post-MTU15 the period is a 15-minute slot (4 obs / hour). Both are valid observations of the unit's strategic choice within the relevant clock-hour, so we stack them in the same regression and let the FE structure absorb the time-aggregation difference. The `crit`, `post`, `y25` indicators are defined at the clock-hour level and replicate across the 4 quarters in the post-period. This gives 4× the observations per hour post-reform, yielding tighter SE; SE are clustered at the date level to preserve correct inference under within-day dependence.

Table 15: DDD and DDDD on q_1 (DA-cleared MWh/unit-hour), da15 window.

	(1) DDD		(2) DDDD	
	b	se	b	se
crit*post*y25*piv (β_{1234}, DDDD)			-18.65**	(7.35)
crit*post*y25 (β_7, DDD)	-16.27***	(2.69)	-0.60	(1.77)
crit*post*piv			13.61	(11.04)
crit*y25*piv			4.37	(7.82)
post*y25*piv			18.96	(30.23)
crit*piv			-0.07	(2.28)
post*piv			-11.81	(10.44)
y25*piv			-4.40	(10.81)
crit*post	22.37***	(2.43)	10.95	(9.18)
crit*y25	-2.00	(1.40)	-5.68	(7.15)
post*y25	-12.01***	(3.37)	-27.93	(29.98)
crit	0.00	(.)	0.00	(.)
post	-3.14	(2.47)	6.77	(9.44)
y25	3.98	(2.77)	7.68	(9.75)
Observations	408800		408800	
R^2	0.092		0.092	
Spec	DDD		DDDD	
Cluster	Date		Firm	
Unit FE	Yes		Yes	
Clock-hour FE	Yes		Yes	
DOW FE	Yes		Yes	

Table 16: DDD and DDDD on q_1 (DA-cleared MWh/unit-hour), ida15 window.

	(1) DDD		(2) DDDD	
	b	se	b	se
crit*post*y25*piv (β_{1234}, DDDD)			-4.01	(6.55)
crit*post*y25 (β_7, DDD)	-9.96***	(3.10)	-6.60	(6.16)
crit*post*piv			-6.86*	(3.07)
crit*y25*piv			3.83	(6.54)
post*y25*piv			14.64	(17.72)
crit*piv			8.01*	(3.58)
post*piv			-2.88	(2.39)
y25*piv			-12.55	(17.89)
crit*post	-6.95***	(1.61)	-1.18*	(0.56)
crit*y25	10.00***	(3.05)	6.79	(6.12)
post*y25	-7.18***	(2.49)	-19.48	(17.61)
crit	0.00	(.)	0.00	(.)
post	-1.11	(0.94)	1.31	(2.35)
y25	7.10***	(2.28)	17.65	(17.82)
Observations	196700		196700	
R^2	0.072		0.073	
Spec	DDD		DDDD	
Cluster	Date		Firm	
Unit FE	Yes		Yes	
Clock-hour FE	Yes		Yes	
DOW FE	Yes		Yes	

Table 17: DDD and DDDD on q_2 (intraday MWh/unit-hour), da15 window.

	(1) DDD		(2) DDDD	
	b	se	b	se
crit*post*y25*piv (β_{1234}, DDDD)			18.78	(15.82)
crit*post*y25 (β_7, DDD)	-1.27**	(0.64)	-17.04	(15.78)
crit*post*piv			-19.42	(15.61)
crit*y25*piv			21.45	(19.14)
post*y25*piv			-2.77	(2.42)
crit*piv			-21.90	(19.30)
post*piv			2.61	(2.55)
y25*piv			-4.20***	(1.11)
crit*post	1.79***	(0.62)	18.10	(15.57)
crit*y25	-2.11***	(0.40)	-20.12	(19.14)
post*y25	-2.96***	(0.71)	-0.63	(0.76)
crit	0.00	(.)	0.00	(.)
post	2.51***	(0.69)	0.32	(0.91)
y25	-4.30***	(0.47)	-0.77	(0.78)
Observations	408800		408800	
R^2	0.105		0.133	
Spec	DDD		DDDD	
Cluster	Date		Firm	
Unit FE	Yes		Yes	
Clock-hour FE	Yes		Yes	
DOW FE	Yes		Yes	

Table 18: DDD and DDDD on q_2 (intraday MWh/unit-hour), ida15 window.

	(1) DDD		(2) DDDD	
	b	se	b	se
crit*post*y25*piv (β_{1234}, DDDD)			11.10	(10.81)
crit*post*y25 (β_7, DDD)	-1.70**	(0.85)	-11.03	(10.19)
crit*post*piv			24.79	(21.64)
crit*y25*piv			-7.27	(8.63)
post*y25*piv			1.46	(4.09)
crit*piv			-30.97	(26.45)
post*piv			-5.40*	(2.85)
y25*piv			-3.39	(4.50)
crit*post	-1.81***	(0.63)	-22.63	(21.42)
crit*y25	0.38	(0.73)	6.49	(7.70)
post*y25	-1.95**	(0.80)	-3.17	(2.16)
crit	0.00	(.)	0.00	(.)
post	-4.77***	(0.60)	-0.24	(0.15)
y25	0.55	(0.65)	3.40	(2.34)
Observations	196700		196700	
R^2	0.162		0.184	
Spec	DDD		DDDD	
Cluster	Date		Firm	
Unit FE	Yes		Yes	
Clock-hour FE	Yes		Yes	
DOW FE	Yes		Yes	

The within-hour quarter index is not used as a treatment-defining variable: ex ante there is no reason firms should bid more aggressively in a particular within-hour quarter regardless of clock-hour (e.g., they do not systematically bid less aggressively in the first quarter of every hour). The granularity is exploited as residual heterogeneity that the FE absorb, not as a treatment dimension. Post-reform cross-section descriptive use of the 15-min data (the D_w, Δp, Δq histograms in the descriptive companion) is a separate complementary exercise.

Supporting evidence elsewhere in the project. See the descriptive-facts sibling document for: (i) RT2 channel characterisation (Part B: REE post-clearing CCGT inclusion, Path A: +1 044 GWh/mo CCGT post-blackout); (ii) zonal concentration (Part B: GN-dominated Cataluña/Levante/Sur, GE-Galicia, IB-Centro); (iii) GN's 840 EUR/MWh bid plateau invariant across regime (Part B: DA bid prices on RT2-up days); (iv) CNMC 2019 sanction (Part B: Cournot reading + CNMC); (v) q_1 collapse same-cal-month CCGT-only with parallel-trends on the post-clearing gap (Part A: q_1 drop).